

Generating Textual Stories from Short Video Clips Subtitle

Computer Vision

Instructor: Professor Marini

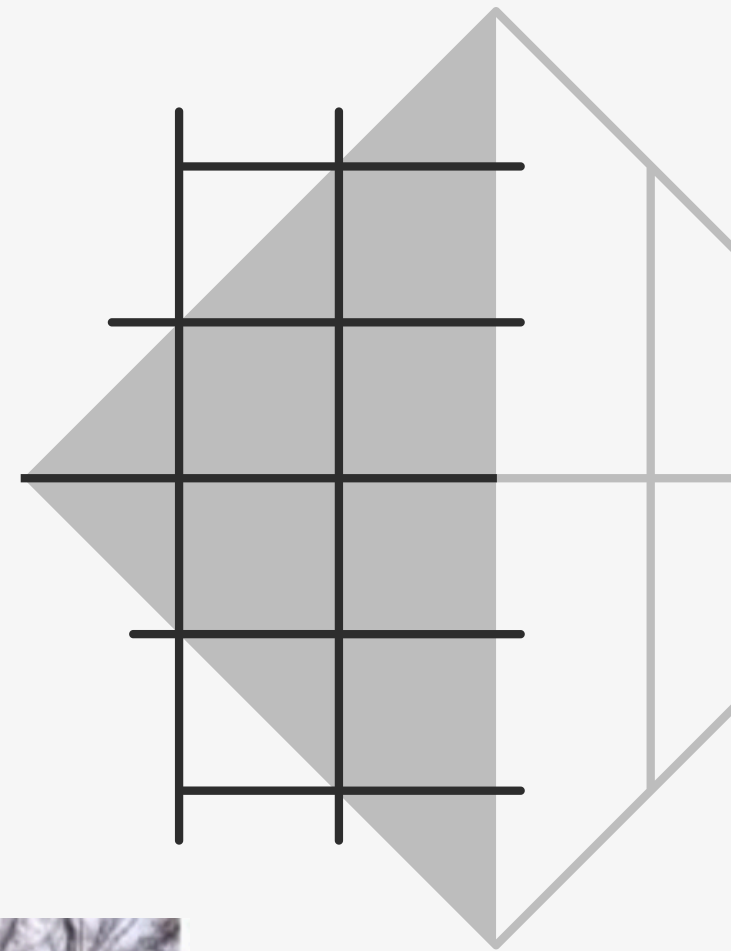
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The Core Problem: Temporal Narrative

- The Challenge: Video storytelling requires bridging the gap between disconnected visual frames and a continuous, logically flowing narrative.
- Current Limitations: Standard models often treat images independently, leading to temporal hallucinations or a failure to connect events chronologically.
- The Goal: Extract discrete visual events and synthesize them into a grammatically cinematic, context-aware story.



Proposed Architecture - A Modular Pipeline

- Step 1: Frame Extraction (Converting clips into keyframes).
- Step 2: Visual Captioning (BLIP-2) (Converting pixels to discrete text).
- Step 3: Narrative Synthesis (LLaMA 3.3) (Weaving captions into a story).
- Architectural Choice: "Late Fusion" (Textual Synthesis) rather than "Early Fusion" (Embedding Aggregation)



Iterative Refinement (V1 to V3)

- **V1: Base Zero-Shot (O1_no_shot)**
 - Issue: High hallucination rate (e.g., describing the camera or video quality).
- **V2: Few-Shot Constraints (O2_few_shots)**
 - Fix: Added strict system prompts forbidding meta-descriptions and provided few-shot examples to enforce physical action descriptions.
- **V3: The "ImageChain" Upgrade – Context-Aware Memory (O3_context_provided)**
 - Evolution: Passing the generated story of Scene 1 as context for Scene 2.

Aligning with the 2025 "ImageChain" Methodology

- Recent Benchmark: IMAGECHAIN: Advancing Sequential Image-to-Text Reasoning (Feb 2025).
- Their Approach: End-to-end Multimodal LLM simulating a multi-turn conversation.
- My Approach: Simulating this exact multi-turn conversation using a lightweight, two-step modular pipeline.
- The Trade-off: Computational efficiency vs. Visual Semantic depth.

IMAGECHAIN: Advancing Sequential Image-to-Text Reasoning in Multimodal Large Language Models

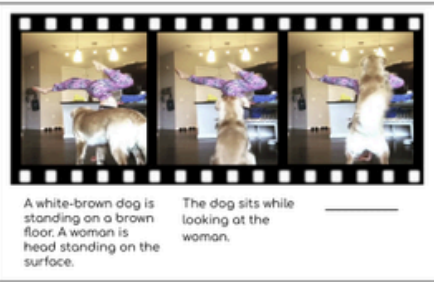
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Abstract

Reasoning over sequences of images remains a challenge for multimodal large language models (MLLMs). While recent models incorporate multi-image data during pre-training, they still struggle to recognize sequential structures, often treating images independently. This work introduces IMAGECHAIN, a framework that enhances MLLMs with sequential reasoning capabilities over image data by modeling visual sequences as a multi-turn conversation. In IMAGECHAIN, images are interleaved with corresponding textual descriptions to form a controlled dialogue that explicitly captures temporal dependencies and narrative progression. Our method optimizes for the task of *next-scene description*, where the model generates a context-aware description of an upcoming scene based on preceding visual and textual cues. We demonstrate that our approach improves performance on the *next-scene description* task – achieving an average improvement from 3.7% to 19% in SimRate, a metric that quantifies semantic similarity to human-annotated ground truths. Moreover, IMAGECHAIN achieves robust zero-shot out-of-domain performance in applications ranging from comics to robotics. Extensive experiments validate that instruction-tuning in a multimodal, multi-turn conversation design is key to bridging the gap between static image understanding and temporally-aware reasoning.¹



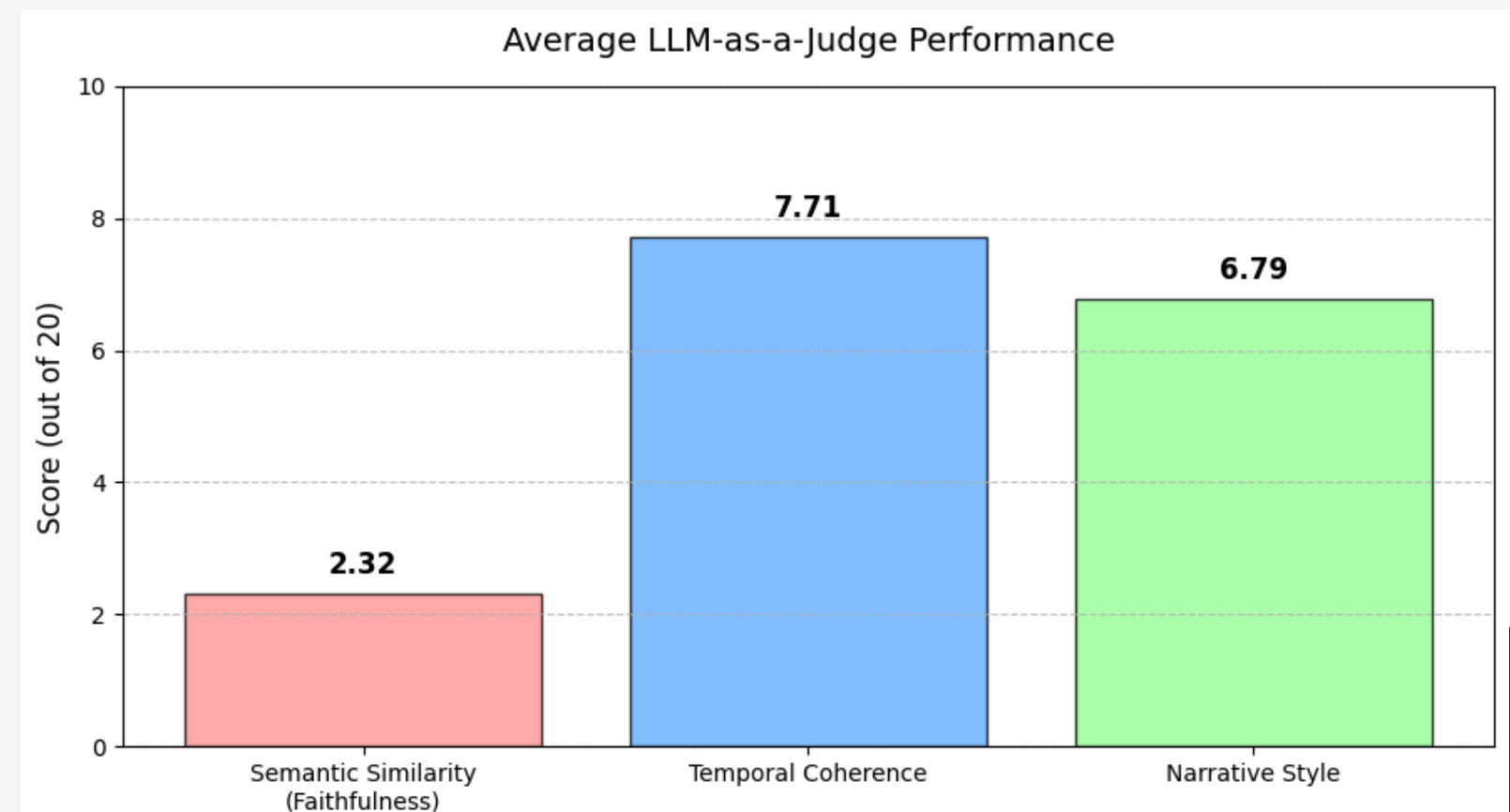
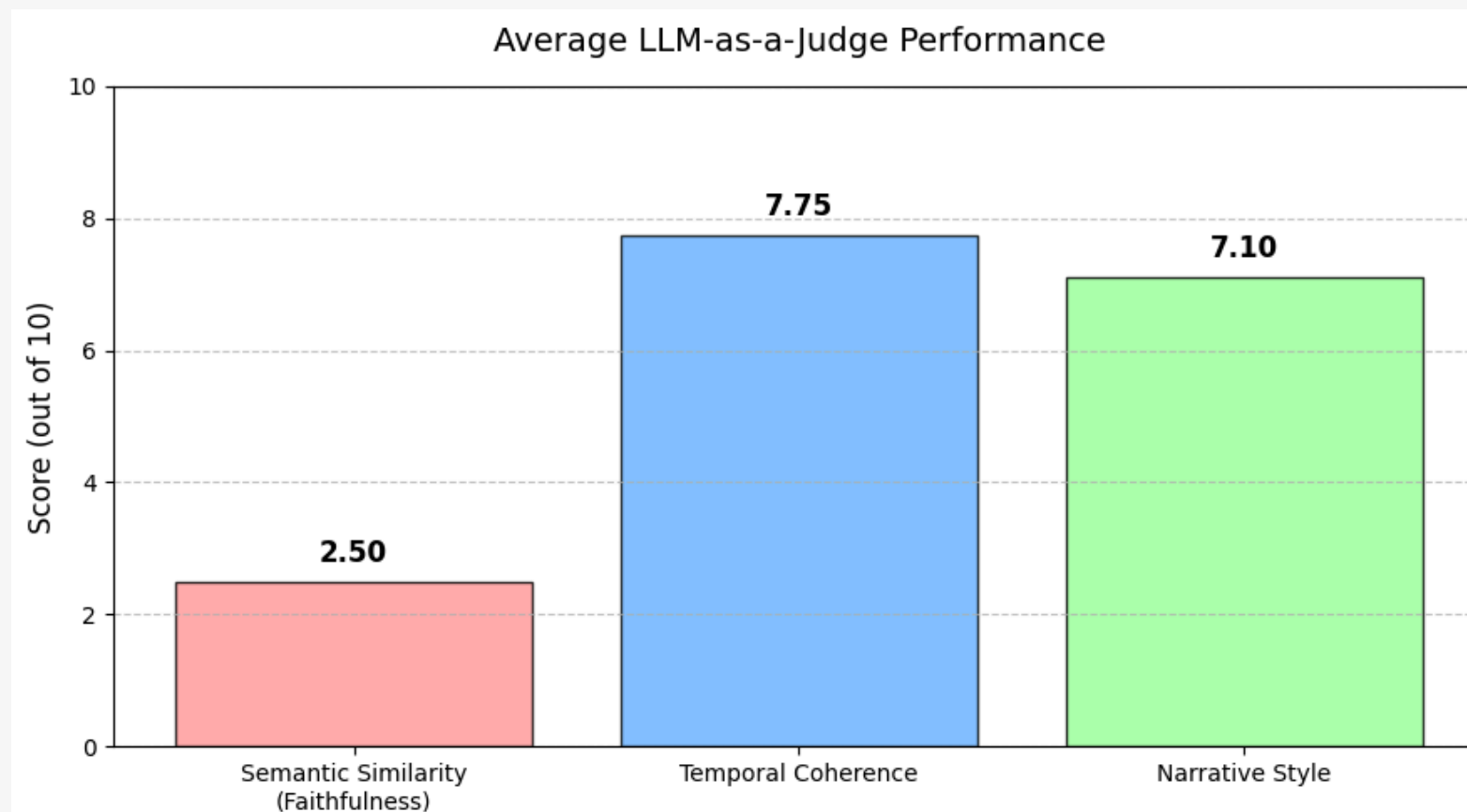
Ground Truth	The dog climbs on a woman and stands.	
Similarity	Does the predicted caption convey a similar meaning to the ground truth caption (Yes or No)?:	
LLaVA-NeXT	The dog moves backward while looking at the woman.	No ☒
Standard Fine-Tuning	The dog moves backward when the woman falls down.	No ☒
IMAGECHAIN (Ours)	The dog stands on its hind legs while looking at the woman.	Yes ☒

Figure 1: Illustration of the proposed *next-scene description* task. Models and techniques that process images independently, such as LLaVA-NeXT and standard fine-tuning, fail to capture the correct progression, leading to errors. In contrast, IMAGECHAIN explicitly models sequences as multi-turn conversations, enabling it to generate a more accurate description, closely aligning with the human-annotated ground truth.

captioning and visual question answering have driven significant progress in recognizing subjects, attributes, and their relationships in individual images (Stefanini et al., 2022; S

Quantitative Evaluation (LLM-as-a-Judge)

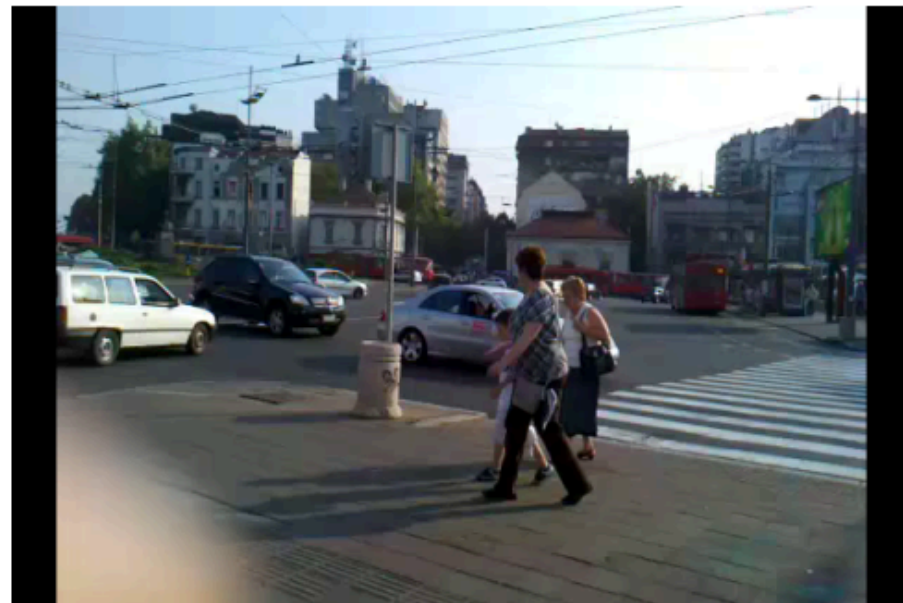
- Metric: Avoided outdated metrics (BLEU, ROUGE) in favor of LLM-as-a-Judge.
- SimRate (Semantic Similarity): Faithfulness to human ground truth.
- Coherence: Logical and chronological flow.
- Style: Natural, observational writing.
- Large Scale Validation: Evaluated on a larger dataset subset (V4) to ensure statistical robustness.



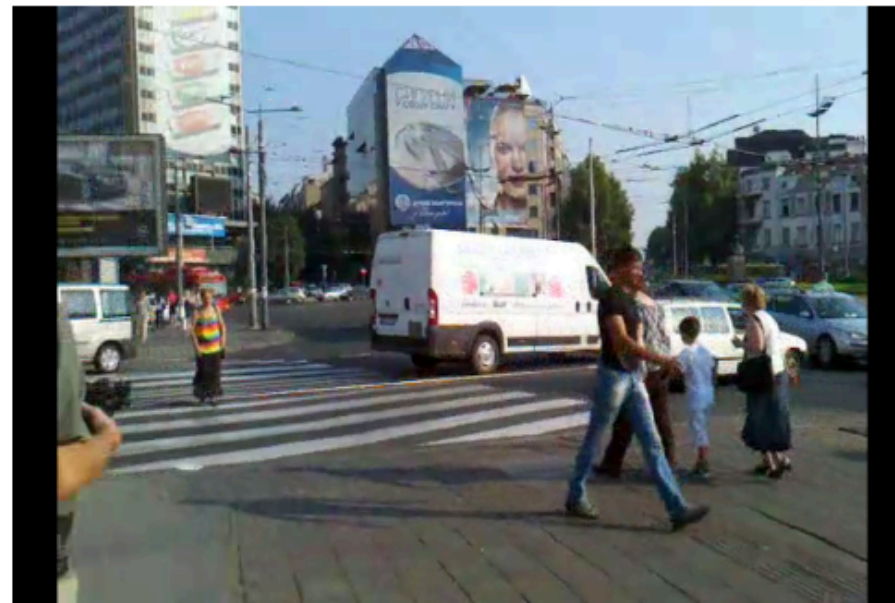
Qualitative Results (In Action)

Original Video Frames: 12644997@N04_5946713307_26456d9fab.3gp (Scene 1)

Frame 1



Frame 2



Frame 8



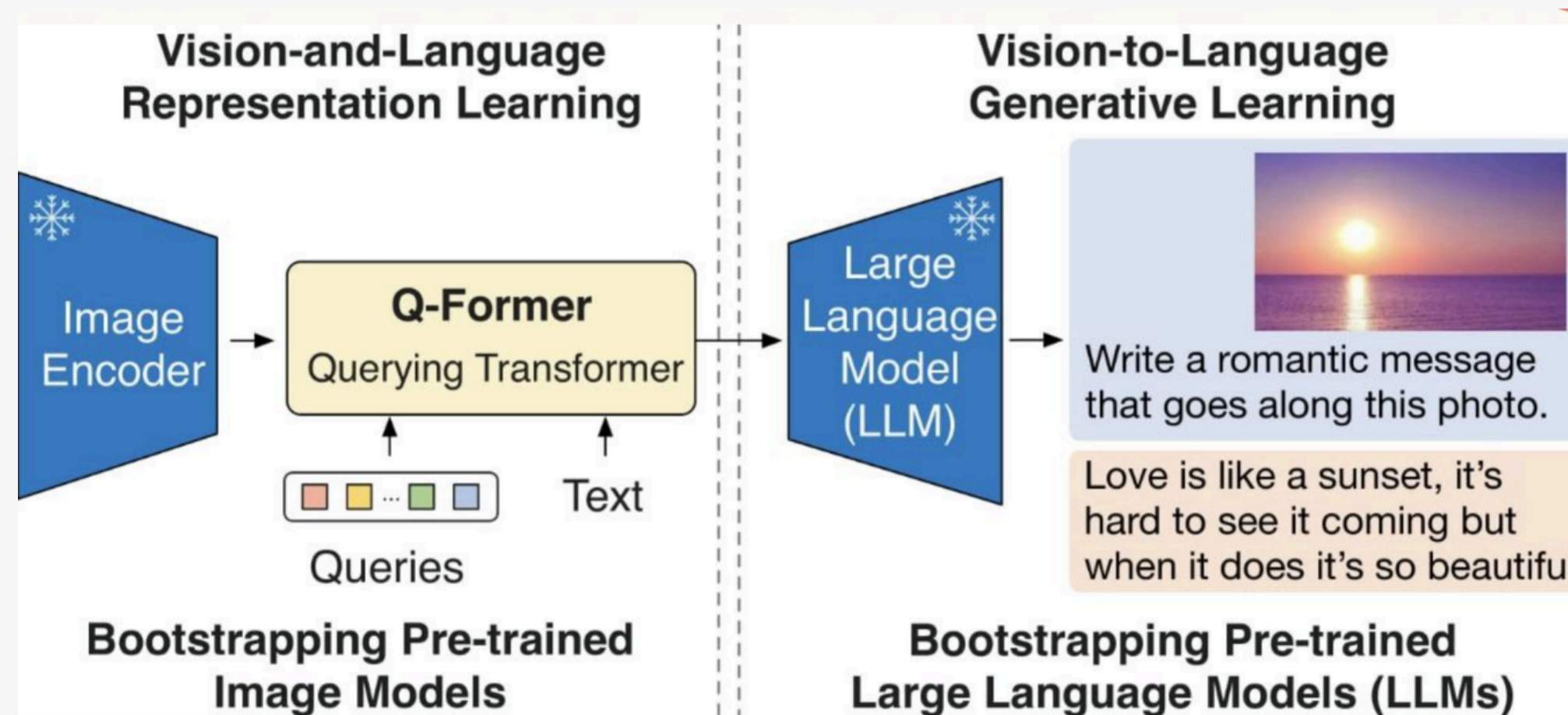
Ground Truth: "A group of people are standing and walking around the street while a group of vehicles are also moving around. The camera pans left and right again and again."

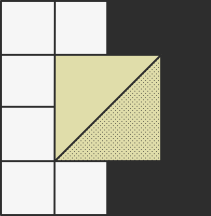
Generated Story:

- Scene1: "A couple of people are walking across the street, their footsteps echoing off the surrounding buildings..."
- Scene2: "A person stands in the middle of a city, flying a kite... A man walks down the sidewalk, a skateboard tucked under his arm... another bus, this one red, drives down the street..."

The Information Gap & Limitations

- The SimRate Discrepancy: Why are Coherence scores high (8–9/10) but Semantic Similarity scores lower (1–6/10)?
- The BLIP-2 Bottleneck: The narrative LLM is "blind." It only knows what the captioner extracts.
- Human vs. AI annotation: The judge penalizes the AI for missing nuanced context that humans naturally infer but a frozen image-captioner misses.





Conclusion & Future Work

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- Conclusion: A modular, context-aware pipeline successfully generates highly coherent temporal narratives without the need for expensive multimodal fine-tuning.
 - Future Work 1: Upgrading the visual encoder (e.g., replacing BLIP-2 with a stronger VLM) to close the SimRate gap.
 - Future Work 2: Real-Time Local Processing. Integrating an OpenCV frame-extractor to allow live processing of local .mp4 video files.

